How many of the above statements are correct?
(a) Only one (b) Only two (c) All three (d) None

Ans. (c) : A thermophile is an organism, a type of extremophile, that thrives at relatively high temperatures between 41 and 122°C. One thermophile, Pyrdobus Fumarii, has been found at temperatures at 113°C which is hotter than boiling point of water. Psychrophiles are organisms capable of growth and reproduction in cold temperatures and are often xerophilic and halophilic as well. The reported minimum growth of microorganism varying from 10°C to –10°C. But for practical cases lower limit of growth for bacteria, yeast or moulds can be taken as –7°C.
Acidophiles are microorganisms that show optimal growth in highly acidic environments. Extreme Citophiles dwell in environments with a pH value less than 3. Thus all the statements are correct.

29. Which one of the following makes a tool with a stick to scrape insects from a hole in a tree or a log of wood?
(a) Fishing cat (b) Orangutan
(c) Otter (d) Sloth bear

Ans. (b) : In a recent study of five captive Sumatran Orangutans it was found that they frequently stripped leaves and twigs from a branch provided and then inserted the tool into a hole to obtain foodstuffs. When they were unable to insert a tool into a hole, the usually modified the tool and changed their tool using technique, such as changing how they grasped the tool.

30. Consider the following :
1. Aerosols 2. Foam agents
3. Fire retardants 4. Lubricants
In the making of how many of the above are hydrofluorocarbons used?
(a) Only one (b) Only two (c) All three (d) All four

Ans. (d) : HFC's are entirely man-made. They are primarily produced for use in refrigeration, air-conditioning, insulating foams and aerosol propellants, with minor uses as solvents and for fire protection.
Lubricants also contain aerosols.

31. Consider the following statements :
1. Jhelum River passes through Wular Lake.
2. Krishna River directly feeds Kolleru Lake.
3. Meandering of Gandak River formed Kanwar Lake.
How many of the statements given above are correct?
(a) Only one (b) Only two (c) All three (d) None

Ans. (b) : The Jhelum river rises from a spring at Verinag situated at the foot of Pir Panjal in the south-eastern part of valley of Kashmir. It flows through Srinagar and the Wular lake before entering Pakistan through a deep narrow gorge.
The Kolleru lake is fed directly by seasonal Budameru and Tammilmeru Streams. It is connected to the Krishna and Gotavari systems by over 68 inflowing drains and channels but not directly Fed.
Kanwar lake in Bihar is residual oxbow take in Bihar formed due in meandering of Gandak river.
Thus statements 1 & 3 are correct and statement is incorrect.

32. Consider the following pairs :

	Port		Well known as
1.	Kamarajar Port	:	First major port in India registered as a company
2.	Mundra Port	:	Largest privately owned port in India
3.	Visakhapatnam Port	:	Largest container port in India

How many of the above pairs are correctly matched?
(a) Only one pair (b) Only two pairs
(c) All three pairs (d) None of the pairs

Ans. (b) : Kamrajar Port is located on the coromandel coast about 24 km. north of Chennai Port. It is the First Port in India which is public company. The Kamrajar Port is the only corporatized major port is registered as a company. Mundra port is the largest Privately owned port in India. It is owned by Adani group. Jawaharlal Nehru Port Trust Nhava Sheva is the largest container port in India. Thus pair 1 & Part 2 are correctly matched.

33. Consider the following trees :
1. Jackfruit (Artocarpus heterophyllus)
2. Mahua (Madhuca indica)
3. Teak (Tectona grandia)
How many of the above are deciduous trees?
(a) Only one (b) Only two (c) All three (d) None

Ans. (b) : Jackfruit is a tropical evergreen tree while Mahua and Teak are tropical deciduous free. Thus (b) is correct.

34. Consider the following statements :
1. India has more arable area than China.
2. The proportion of irrigated area is more in India as compared to China.
3. The average productivity per hectare in Indian agriculture is higher than that in China.
How many of the above statements are correct?
(a) Only one (b) Only two (c) All three (d) None

Ans. (b) : India's arable land area is 179.8 million hectares while China's arable land area is 165.2 million hectares.
India's proportion of irrigated area is 48% while China's proportion of irrigated area is 41%. Thus proportion of irrigated area in India is more as compared to China.
Agricultural productivity in China is almost 50 to 100% more than that of India. Thus Statements (1 & 2) are correct.

35. Which one of the following is the best example of repeated falls in sea level, giving rise to present-day extensive marshland?
(a) Bhitarkanika Mangroves
(b) Marakkanam Salt Pans
(c) Naupada Swamp
(d) Rann of Kutch

Ans. (d) : The Rann of Kutch was once shallow part of Arabian Sea, Until a geological shift closed off the connection with sea. The region became a seasonal marshy salt desert over the years. During Monsoons the region fills up with water and forms a wetland. In the summer the water dries to create a bed of white salty land. The Great Rann is a bowl shaped depression, its eastern edge merges into the drainage area of the Luni river.

36. Ilmenite and rutile, abundantly available in certain coastal tracts of India, are rich sources of which one of the following?
(a) Aluminium (b) Copper (c) Iron (d) Titanium

Ans. (d) : Ilmenite and Rutile are both minerals that contain high concentration of titanium.
In India certain coastal trails are known for significant ilnenite and rutile deposits. These are found in beach deposits and are extracted through beach sand mining operations.
The extracted ilnenite and rutile are processed to obtain titanium dioxide which is then further refined to produce metallic titanium.

37. About three-fourths of world's cobalt, a metal required for, the manufacture of batteries for electric motor vehicles, is produced by
(a) Argentina
(b) Botswana
(c) the Democratic Republic of the Congo
(d) Kazakhstan

Ans. (c) : The Democratic Republic of Congo is the world's largest producer of cobalt accounting for about three-fourths of the global protection. The country reported output of 1,30,000 MT in 2022. Cobalt is crucial for manufacture of some electronic products (Smartphones & Laptops) and batteries of Electric Vehicles.

38. Which one of the following is a part of the Congo Basin?
(a) Cameroon (b) Nigeria (c) South Sudan (d) Uganda

Ans. (a) : The Congo Basin region in Central Africa includes multiple countries which are as follows :
- Cameroon
- Central African Republic
- Democratic Republic of Congo
- Republic of Congo
- Equatorial Guinea
- Gabon

Thus 'Cameroon' is the correct answer.

39. Consider the following statements :
1. Amarkantak Hills are at the confluence of Vindhya and Sahyadri Ranges.
2. Biligirirangan Hills constitute the easternmost part of Satpura Range.
3. Seshachalam Hills constitute the southernmost part of Western Ghats.
How many of the statements given above the correct?
(a) Only one (b) Only two (c) All three (d) None

Ans. (d) : Amarkantak is at confluence of Vindhya and Satpura Range.
Annamalai hills and Cardmom hills are the southernmost point of Western Ghats.
Sheshachalam Hills are part of eastern Ghats.
Thus none the statements are correct.

40. With reference to India's projects on connectivity, consider the following statements:
1. East-West Corridor under Golden Quadrilateral Project connects Dibrugarh and Surat.
2. Trilateral Highway connects Moreh in Manipur and Chiang Mai in Thailand via Myanmar.
3. Bangladesh-China-India-Myanmar Economic Corridor connects Varanasi in Uttar Pradesh with Kunming in China.
How many of the above statements are correct?
(a) Only one (b) Only two (c) All three (d) None

Ans. (d) : The East-West corridor is a part of National Highways Development Project which aims to connect Silchar in Assam to Porbandar in Gujarat.
The Trilateral Highway is a connectivity project that aims to connect Moreh in Manipur, India with Maesot in Thailand via Myanmar.
The Bangladesh-China-India-Myanmar (BCIM) is a proposed economic corridor that aims to connect Kunming in China with Kolkata in India via Myanmar and Bangladesh.
Thus all the statements are incorrect.

41. Consider the following statements :
Statement-I : Interest income from the deposits in Infrastructure Investment Trusts (InvITs) distributed to their investors is exempted from tax, but the dividend is taxable.
Statement II : InvITs are recognized as borrowers under the Securitisation and Reconstruction of Financial Assets and Enforcement of Security Interest Act, 2002'.
Which one of the following is correct in respect of the above statements :
(a) Both Statement-I and Statement-II are correct and Statement-II is the correct explanation for Statement-I.
(b) Both Statement-I and Statement-II are correct and Statement-II is not the correct explanation for Statement-I.
(c) Statement-I is correct but Statement-II is incorrect.
(d) Statement-I is incorrect but Statement-II is correct.

Ans. (d) : InvITs are investment vehicle that pool Funds From investors to invest in infrastructure projects. Latest budget 2023-24 made the provision that all income distributed by InvITs, including interest income, dividend income and rental income will be taxable.
InvITs are classified as borrowers under the securitization and Reconstruction of Financial Assets and Enforcement of Securities Interest Act, 2002.
Thus statement (1) is incorrect and statement 2 is correct.

42. Consider the following statements :
Statement-I : In the post-pandemic recent past, many Central Banks worldwide had carried out interest rate hikes.
Statement II : Central Banks generally assume that they have the ability to counteract the rising consumer prices via monetary policy means.
Which one of the following is correct in respect of the above statements :
(a) Both Statement-I and Statement-II are correct and Statement-II is the correct explanation for Statement-I.
(b) Both Statement-I and Statement-II are correct and Statement-II is not the correct explanation for Statement-I.
(c) Statement-I is correct but Statement-II is incorrect.
(d) Statement-I is incorrect but Statement-II is correct.

Ans. (a) : In the post-pandemic recent past, many central banks worldwide have indeed carried out interest rate hikes central banks use interest rate adjustments as one of their monetary policy tools to manage the economy.